

## Project Initialization and Planning Phase

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| Date          | 15 March 2024                                                                                           |
| Team ID       | PNT2022TMID124356                                                                                       |
| Project Title | Smart agriculture production optimization engine- Applicant<br>Credibility Prediction for Loan Approval |
| Maximum Marks | 5 Marks                                                                                                 |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

| Project Overview  |                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to revolutionize the loan approval process by implementing advanced machine learning techniques, ensuring faster and more accurate assessments. |
| Scope             | The project comprehensively assesses and enhances the loan approval process, incorporating machine learning for a more robust and efficient system.                      |
| Problem Statement |                                                                                                                                                                          |

Description Addressing inaccuracies and inefficiencies in the current loan approval system adversely affects operational efficiency and customer satisfaction.

Impact Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success.

### Proposed Solution

Approach Employing machine learning techniques to analyze and predict creditworthiness, creating a dynamic and adaptable loan approval system.

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|--------------|-----------------------------------------------------------------------|
| Key Features | - Implementation of a machine learning-based credit assessment model. |
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|  | - Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes. |
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### Resource Requirements

| Resource Type       | Description                             | Specification/Allocation |
|---------------------|-----------------------------------------|--------------------------|
| <b>Hardware</b>     |                                         |                          |
| Computing Resources | CPU/GPU specifications, number of cores | T4 GPU                   |
| Memory              | RAM specifications                      | 8 GB                     |

|                                                                                                                                                                                             |                                       |                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------|
| Storage                                                                                                                                                                                     | Disk space for data, models, and logs | 1 TB SSD                                       |
| <b>Software</b>                                                                                                                                                                             |                                       |                                                |
| Frameworks Python frameworks Flask<br><br>Libraries Additional libraries scikit-learn, pandas, numpy, matplotlib, seaborn Development Environment IDE Jupyter Notebook, pycharm <b>Data</b> |                                       |                                                |
| Data                                                                                                                                                                                        | Source, size, format                  | Kaggle dataset, 614, csv UCI dataset, 690, csv |